PHPD — Production Server Specs (One Page)

Generated: 2026-04-27 | Profile: missioncritical, 100–300 concurrent users, mixed access, uploads 50–200 GB/month, retain forever

1) Application stack (from repo)

- Backend: Django + DRF + JWT (SimpleJWT), GeoDjango + REST framework GIS.
- Database: PostgreSQL + PostGIS (GeoDjango postgis engine used in local/production settings).
- Frontend: React (Vite). Production deploy is static build + API base URL via Vite env.
- Uploads: images, documents, XER files; media must be shared across app servers in production.

2) Recommended production architecture (hostingneutral)

- Edge: TLS termination + reverse proxy/load balancer (HA pair). Only 443 exposed publicly.
- App tier: 2+ Django app nodes behind LB (Nginx + Gunicorn/Uvicorn). Scale horizontally.
- DB tier: PostgreSQL + PostGIS with HA (primary + standby). DB is private (no public access).
- Media storage (best choice): object storage (S3compatible/cloud blob) + optional CDN for downloads.
- Backups/DR: PITR for DB; object versioning + replication for media; regular restore tests.

3) Initial sizing (can scale up/down)

- Load balancer / reverse proxy (x2): 2 vCPU, 4 GB RAM, 30–50 GB disk.
- App servers (x2 start): 8 vCPU, 16–32 GB RAM, 80–150 GB disk; stateless (no permanent media).
- DB servers (x2): 8–16 vCPU, 32–64 GB RAM, NVMe/SSD; start 1–2 TB data + 200–500 GB WAL/logs.
- Object storage: capacity plan ~0.6–2.4 TB/year (based on 50–200 GB/month) + replication overhead.

4) Security, networking, and ops

- Network segmentation: public LB only; app/DB on private subnets; restrict DB to app security group only.
- Django hardening: DEBUG=False, strict ALLOWED_HOSTS, secure cookies, rotate secrets via vault/secret manager.
- Observability: centralized logs; metrics/alerts for app saturation, DB replication lag, slow queries, disk usage.
- SLO focus: remove single points of failure; planned maintenance windows; patching and backup verification.

Note: Hosting platform not finalized; this spec is vendorneutral and HAoriented.